Heurion 2.0: A Contract-Guarded Dual-Surface Architecture with Self-Evolving Memory for Citation-Grounded Clinical Research Authoring

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Abstract

Autonomous large language model (LLM) agents often edit documents via shell scripts and post-hoc file checks. In clinical research authoring, this workflow risks corrupting Office Open XML (`.docx`/`.pptx`) formatting, overwriting concurrent human edits, inserting unverified bibliographic IDs (DOIs/PMIDs), exposing Protected Health Information (PHI), and repeating guard failures across sessions.

We present **Heurion 2.0**, a clinical authoring platform combining a *Contract-Guarded Dual-Surface Architecture* with a *Three-Layer Self-Evolving Memory System*. A shared CRDT core (ProseMirror + Yjs) with stable block IDs projects a rich-text editor for clinicians and a guarded Annotated Markdown surface for AI agents with per-turn undo and original-slice OOXML export. Pre-write guards block unregistered citations, a two-stage neuro-symbolic PHI funnel pairs sub-millisecond checksum/regex validators with a pre-masked local 1.5B small language model (`qwen2.5:1.5b`) and deterministic biomedical output filtering, and runtime guard/revert signals feed a utility-ranked procedural memory store isolated from declarative preferences. Across 10,839 live LLM turns spanning 20 open-access PMC clinical manuscripts and 1,010 deterministic/local-SLM guard evaluation records under pre-specified protocols (`ANALYSIS_PLAN.md` and `ANALYSIS_PLAN_v2.md`), Heurion 2.0 demonstrated: (1) in a confirmatory 2,340-turn three-arm manuscript co-editing benchmark across 10 new 2026 PMC manuscripts (following a 2,148-turn development phase), a turn rework rate of 4.9% (38/780; 2.1% [16/780] after excluding evaluator false positives, with 0 lost clinician edits) compared to 14.0% (109/780, OR = 32.47, Holm $p = 1.03 \times 10^{-16}$) for direct file scripting (`B1`) and 10.3% (80/780, OR = 9.66, Holm $p = 1.23 \times 10^{-8}$) for tool-assisted file scripting (`B2`), while reducing per-turn API cost by 30%–33% (GMR = 0.70 and 0.67, Holm $p < 10^{-72}$); (2) on a locked 100-note Chinese clinical de-identification test set (745 PHI spans: 485 narrative, 260 labeled) and 200 hard-negative clinical sentences, an increase in overall PHI recall from 27.4% (204/745, Stage-1 regex alone; 3.5% on narrative spans) to 87.0% (648/745; 83.3% [404/485] on narrative spans and 93.8% [244/260] on labeled spans) with 97.6% (482/494) precision and a 6.0% (12/200) hard-negative false-positive rate (vs. 92.0% [184/200] for the unvalidated SLM and 93.5% [187/200] for exhaustive rules); and (3) in a confirmatory 4,608-turn, 4-arm (`off`, `frozen`, `placebo`, `utility`) × 16-user × 8-session (6 training + 2 held-out) longitudinal ablation on uncovered structural constraints, a 63.7% reduction in pre-write guard rejections per turn (0.285 vs. 0.784 [`off`], 0.774 [`frozen`], and 0.771 [`placebo`]; held-out Sessions 7–8: 0.194 vs. 0.812, 0.802, 0.781), with negative binomial mixed-effects modeling (`glmmTMB`) confirming highly significant advantages over `off` (RR = 0.363, Holm $p = 2.64 \times 10^{-55}$), static rules `frozen` (RR = 0.368, Holm $p = 3.87 \times 10^{-54}$), and length-matched `placebo` (RR = 0.369, $p = 1.32 \times 10^{-53}$) alongside a +11.72% gain in task completion (92.8% vs. 81.1%; 100.0% [288/288] in held-out sessions).

Keywords: Heurion 2.0, Collaborative AI Agents, Clinical Research Authoring, Yjs CRDT

Co-Editing, Office Open XML (`.docx`/`.pptx`) Preservation, Model Context Protocol (MCP), Citation Hallucination Prevention, PHI Redaction, Self-Evolving Agent Memory, Human-AI Permission Symmetry

Statement of Significance

Problem: Autonomous LLM agents that edit clinical manuscripts via file scripts risk corrupting journal OOXML formatting, overwriting concurrent clinician edits, leaking narrative Protected Health Information (PHI) to cloud APIs, and repeating operational guard violations across sessions.

What is Already Known: Deterministic regex scrubbers miss unstructured narrative PHI, whereas unvalidated small language models suffer from severe clinical false positives ($> 90\%$). Static system-prompt rules cannot anticipate evolving collaborator edits or uncovered document constraints.

What This Paper Adds: Heurion 2.0 introduces (i) a contract-guarded Yjs CRDT co-editing engine that cuts manuscript rework by $3.2\times$–$9.7\times$ in odds (OR = 9.66–32.47) and lowers cost by 30%–33% over script agents; (ii) a local two-stage PHI funnel achieving 87.0% overall (83.3% narrative) recall at 97.6% precision and 6.0% hard-negative FPR; and (iii) an online procedural rule crystallization engine that reduces guard rejections by 63.7% (RR = 0.363–0.369, $p < 10^{-53}$) and achieves 100% held-out task completion across 4,608 longitudinal turns.

Highlights:

- Dual-surface CRDT core preserves OOXML slices and blocks concurrent human-edit loss.
- Confirmatory 2,340-turn benchmark cuts rework (OR = 9.66–32.47) and cost by 30–33%.
- Local 1.5B SLM + symbolic validator reaches 83.3% narrative PHI recall at 6.0% FPR.
- Actionable procedural rule crystallization cuts guard rejections by 63.7% (4,608 turns).
- Four-arm design isolates online learning from static rules and prompt-length placebo.

1 Introduction

Clinical and biomedical research workflows require high standards of traceability, structural fidelity, and regulatory compliance. A single clinical study typically spans patient longitudinal record curation, cohort screening, biostatistical modeling (e.g., baseline Table 1 generation, Kaplan–Meier survival curves, Cox proportional hazards regression, CONSORT flow diagrams), and the iterative authoring of peer-reviewed manuscripts (`.docx`) and academic presentation decks (`.pptx`) [1, 2]. While recent agentic systems have demonstrated strong capabilities in code generation [3, 4] and open-domain literature synthesis [5, 6], deploying autonomous agents into clinical research environments exposes four architectural barriers:

1. **The Free-Form File Mutation Trap:** Conventional script-based document agents treat `.docx` and `.pptx` files as opaque disk artifacts manipulated via generated Python scripts (e.g., `python-docx`) followed by post-hoc quality checks. In practice, script-based rewrites can strip publisher XML attributes, lose paragraph addressing identifiers (`w14:paraId`) during LibreOffice layout passes, suffer from punctuation/whitespace reproduction drift in full-block replacements, and cannot selectively undo an AI turn once a human co-author makes concurrent edits.
2. **Citation Fabrication and Ungrounded Clinical Claims:** LLMs frequently hallucinate plausible-looking Digital Object Identifiers (DOIs), PubMed IDs (PMIDs), or numerical hazard ratios (HR) and p-values [9, 10, 11]. Post-hoc prompting alone cannot guarantee that every inline reference corresponds to a verified bibliographic record or that statistical assertions match the cited source text.
3. **Patient Privacy (PHI) Exposure and Permission Asymmetry:** Clinical investigators work with raw electronic medical records containing Protected Health Information (patient names, national ID numbers, admission codes, phone numbers) [12, 13]. Passing unredacted records to external LLMs conflicts with health-data protection law in both jurisdictions we

target: China's Personal Information Protection Law (PIPL) classifies medical and health information as sensitive personal information that may be processed only for a specific purpose, with separate consent and strict protective measures, and imposes additional conditions on cross-border transfer [16]; the U.S. HIPAA Privacy Rule restricts disclosure of PHI unless it is de-identified, for example by removing the 18 Safe Harbor identifiers [15]. Conversely, granting agents unrestricted database or shell access creates privilege-escalation and indirect prompt-injection risks [17].

4. **Cross-Session Amnesia and Memory Pollution:** When an agent encounters a structural guard rejection (e.g., attempting to overwrite a human-edited block or inserting an unregistered citation) or when a clinician reverts an AI turn, stateless agents repeat the same operational failure in subsequent turns. Naively appending all historical observations into a single flat memory pool mixes subjective user writing preferences with hard operational constraints, exhausting context budgets with low-utility noise [19, 20].

1.1 Why Script-and-Audit File Mutation Fails in Collaborative Authoring

When general-purpose coding agents operate inside a workspace shell, they typically invoke Python scripts (`python-docx` or XML manipulators) to unpack, mutate, and repackage `.docx` archives on disk, optionally pairing edits with a post-turn audit script that checks text survival rates, paragraph drift, or DOI validity. Collaborative clinical authoring exposes three structural limitations in this "script-and-audit" paradigm:

1. **AI Editing Can Disrupt Document Structure:** When an agent edits a manuscript via ad-hoc scripts or whole-block re-serialization, it risks altering the underlying OOXML structure—stripping journal-template paragraph and run properties (`<w:pPr>`, `<w:rPr>`), breaking list numbering chains and multi-row table merges, detaching inline review comments and track-changes spans, and losing paragraph identifiers (`w14:paraId`) across layout passes.

2. **Risk of Overwriting Concurrent Human Edits:** Because script-based agents read a file snapshot, rewrite whole paragraphs or sections, and write the file back, any concurrent edits made by the clinician mid-turn (such as inserting a clinical caveat in the same paragraph, adjusting a statistic, or attaching a comment) can be overwritten. Reconciling an AI-rewritten file with concurrent browser edits via post-hoc three-way merges is brittle.

3. **Post-Hoc Auditing Cannot Prevent Destructive Writes:** Auditing *after* the write occurs only after mutation: once a script has altered document formatting or overwritten a co-author's edit, a post-hoc check can only flag the discrepancy or trigger a coarse file-level rollback that discards the entire turn. Preventing structural corruption and human-edit loss requires replacing free-form file mutation with a shared, fine-grained collaborative data structure and pre-write structural guards.

1.2 The Heurion 2.0 Architecture and Contributions

Heurion 2.0 shifts from *post-hoc file auditing* to *pre-write structural contracts over a shared CRDT core*, coupled with a *signal-driven self-evolving execution harness*. Our primary contributions are fourfold:

- **Contract-Guarded Dual-Surface CRDT Engine (Section 3):** We design a unified ProseMirror + Yjs document core [7, 8] with persistent block addressing (`{#id}`), node-level optimistic concurrency control (`base_rev`), minimal-diff XML fragment updates that preserve inline marks and comment anchors, per-turn selective CRDT undo (`revertTurn`), and raw-slice OOXML patch export (`node_src`) that writes back original XML bytes for unmodified blocks.

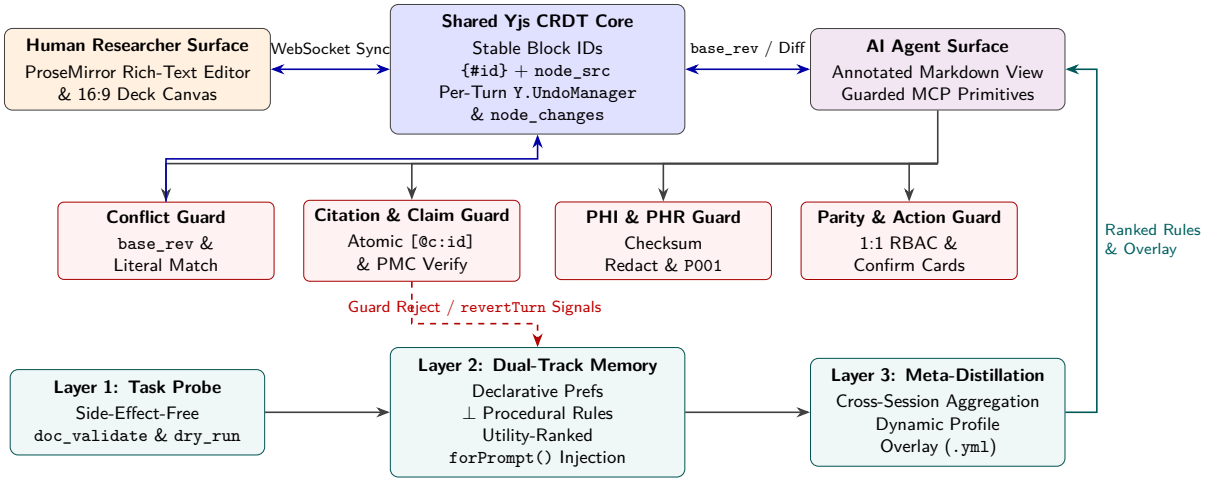


Figure 1: System architecture of Heurion 2.0, a contract-guarded clinical research and scientific authoring platform. Top: Human researchers and AI agents co-edit a shared Yjs CRDT document and slide core through synchronized surface projections (ProseMirror rich-text/deck canvas and Annotated Markdown MCP tools). Middle: All AI mutations pass through four deterministic pre-write structural guards (Conflict Guard, Citation & Claim Guard, PHI & PHR Guard, and Parity & Action Guard) before committing to the CRDT state. Bottom: Runtime guard rejections and human turn reverts feed a three-layer self-evolving memory system that dynamically injects high-utility procedural directives into future turns.

- **Pre-Write Citation and Two-Stage Neuro-Symbolic PHI Guards (Section 4):** We enforce a syntactic pre-write ban on raw inline DOIs/PMIDs, requiring registration via Crossref, PubMed, and OpenAlex into atomic citation nodes (`[@c:id]`), and design a two-stage PHI redaction cascade that pairs sub-millisecond checksum/regex validators with a pre-masked local 1.5B small language model (`qwen2.5:1.5b`) and symbolic output filtering to redact both structured identifiers and unstructured clinical narratives within the local loopback boundary. For system completeness, we also outline our multi-agent claim verification pipeline and sandboxed biostatistical tools (which are not quantitatively evaluated in this study).
- **Three-Layer Self-Evolving Agent Architecture (Section 5):** We introduce a hierarchical adaptation loop comprising (Layer 1) side-effect-free pre-flight validation probes (`doc_validate` / `dry_run`), (Layer 2) a dual-track memory architecture that physically separates human-approved declarative preferences from runtime-crystallized procedural strategies ranked by a cue-based utility score under a fixed prompt budget, and (Layer 3) background meta-strategy distillation into dynamic agent profile overlays.
- **CI-Enforced Human-AI Permission Symmetry (Section 6):** We establish, and mechanically check via an automated parity test over the router table, an architectural invariant across all platform endpoints: every REST capability either maps 1-to-1 to an in-process Model Context Protocol (MCP) [18] tool executing under the user's exact RBAC identity, or is explicitly justified in a restricted exclusion set, with high-risk mutations gated by human-only confirmation cards.

2 System Overview and Design Principles

Figure 1 illustrates the end-to-end architecture of Heurion 2.0. The platform decouples *structured document/slide state mutation* from *sandboxed statistical computation*.

Heurion 2.0 is governed by three formal architectural invariants that enforce pre-write document integrity, human-AI permission symmetry, and memory orthogonality:

Architectural Invariant 1 (Pre-Write Structural Safety)**.** *Let S_r denote the CRDT document state at revision r, and let $\mathcal{O} = \{o_1, \ldots, o_k\}$ be a batch of structured edit operations submitted by an AI turn t against base revision $r_{base} \leq r$. Let $\Delta_{user}(r_{base}, r)$ be the set of block IDs modified by human actors in $(r_{base}, r]$, $\mathcal{P}_{sugg}(S_r)$ be the set of blocks holding unresolved clinician track-changes suggestions, $Anchors(b)$ be the set of open review-comment anchors attached to block b, and $\mathcal{R}_{cite}(S_r)$ be the set of verified citation identifiers registered in the document store. The state transition $\mathcal{T}(S_r, \mathcal{O}, r_{base})$ commits if and only if every $o_i \in \mathcal{O}$ with target block $b = target(o_i)$ satisfies all four pre-write conditions:*

$$\begin{aligned} BlockMut(o_i) &\implies b \notin \Delta_{user}(r_{base}, r) \cup \mathcal{P}_{sugg}(S_r), \\ ReplaceText(o_i) &\implies \big|Matches\big(find(o_i),\, S_r[b]\big)\big| = 1 \;\wedge\; b \notin \mathcal{P}_{sugg}(S_r), \\ Anchors(S_r[b]) &\subseteq Anchors\big(o_i(S_r[b])\big) \cup DropAnchors(o_i), \\ Cites(o_i) &\subseteq \mathcal{R}_{cite}(S_r) \;\wedge\; RawCitePattern(o_i) = \emptyset. \end{aligned} \tag{1}$$

Otherwise, the entire batch $\mathcal{O}$ is atomically rejected prior to mutating S_r, returning a structured diagnostic code (`conflict_user_edited`, `pending_suggestion`, `text_not_found`, `ambiguous_match`, `anchor_has_open_comments`, or `citation_not_registered`) and the current block snapshot.

Architectural Invariant 2 (Human-AI Permission Symmetry)**.** *Let $\mathcal{E}_{REST}$ be the set of all HTTP API routes exposed to the web user interface, and let $\mathcal{E}_{MCP}$ be the set of Model Context Protocol (MCP) tools exposed to the AI agent. Every route $e \in \mathcal{E}_{REST}$ satisfies the disjoint partition:*

$$\mathcal{E}_{REST} = \phi(\mathcal{E}_{MCP}) \sqcup \mathcal{E}_{NOT_FOR_AI}, \tag{2}$$

where $\phi : \mathcal{E}_{MCP} \to \mathcal{E}_{REST}$ is an in-process Loopback dispatcher that executes under the authenticated user's exact session token and tenant isolation context, and $\mathcal{E}_{NOT_FOR_AI}$ is an explicitly audited set of human-only lifecycle endpoints (e.g., credentials, high-risk confirmation approval).

Architectural Invariant 3 (Declarative-Procedural Memory Orthogonality)**.** *Let $\mathcal{M}_D$ denote the declarative memory store (subjective authoring preferences, stylistic conventions, domain terminology) and $\mathcal{M}_P$ denote the procedural harness store (operational guard avoidance rules and subagent delegation policies). Updates to $\mathcal{M}_D$ require explicit human confirmation (status $\in \{$`candidate` $\xrightarrow{human}$ `confirmed`$\}$), whereas $\mathcal{M}_P$ evolves autonomously from deterministic runtime execution signals Σ_{exec} without polluting $\mathcal{M}_D$ or requiring manual administrative curation.*

3 Dual-Surface Co-Editing and Original-Slice Document Engine

How can autonomous AI agents edit complex Word (`.docx`) manuscripts and PowerPoint (`.pptx`) presentation decks without corrupting journal-template formatting or overwriting concurrent human edits? A central architectural observation in Heurion 2.0 is that Office Open XML (`.docx` and `.pptx`) files are delivery formats, not collaborative editing data structures. When an agent edits a `.docx` file on disk via a Python script while a human views the document in a browser, the two actors operate on divergent state representations. Heurion 2.0 eliminates this divergence by replacing direct file scripting with a Contract-Guarded Dual-Surface CRDT Engine [7, 8].

3.1 Shared CRDT Core with Lifecycle-Stable Block Addressing

In Heurion 2.0, every manuscript and presentation deck resides in a server-authoritative Yjs CRDT document (`Y.Doc`) backed by SQLite state snapshots and an append-only revision log (`doc_commits`). Every block-level ProseMirror node (heading, paragraph, list item, table, figure, or slide element) is assigned a short, lifecycle-stable identifier (`attrs.id`, rendered as `{#id}`). When a human splits a paragraph or pastes content in the browser, the server-side `flush()` handler automatically detects duplicate or missing block IDs and assigns fresh unique IDs inside a system CRDT transaction, ensuring the addressing space remains bijective.

From this single CRDT core, Heurion 2.0 projects two synchronized surfaces:

1. **Human Surface:** A real-time collaborative ProseMirror rich-text editor (for manuscripts) and an interactive 16:9 SVG/HTML slide canvas (for decks) synchronized via WebSocket (`y-protocols/sync`).
2. **Agent Surface:** A compact *Annotated Markdown* view generated on demand via `doc_read` and `slide_read`. Each block is suffixed with its stable address `{#id}`, inline citations are rendered as atomic tokens `[@c:cite_id]`, figures as `![alt](asset:id)`, and active inline review comments as `{#c:comment_id}`.

3.2 Node-Level Conflict Guards and Minimal-Diff Fragment Updates

To prevent an AI turn in Heurion 2.0 from overwriting concurrent human edits, every `doc_edit` call specifies the `base_rev` read by the agent. As formalized in Invariant 1, the platform queries the `node_changes` index for all blocks modified by human actors since `base_rev`:

- **Block-Level Structural Operations (`replace_block`, `delete`):** If a target block $b \in \Delta_{\text{user}}(r_{\text{base}}, r)$, the operation is rejected with error code `conflict_user_edited`, returning the human's latest block markdown so the agent can adapt. Similarly, blocks carrying unresolved track-changes suggestions ($b \in \mathcal{P}_{\text{sugg}}(S_r)$) are protected by the `pending_suggestion` guard, and blocks holding open review comments require either preserving the comment anchors or explicitly declaring `drop_anchors` (`anchor_has_open_comments`).
- **Substring Replacement (`replace_text`):** For surgical phrase edits within a paragraph, the literal `find` substring acts as its own optimistic concurrency guard: if the human edited a different sentence in the same paragraph, `find` still matches uniquely and succeeds; if the human altered the target phrase itself or if `find` matches multiple occurrences, validation fails cleanly with `text_not_found` or `ambiguous_match`.
- **Mark-Preserving Minimal Diffs:** Notably, `replace_text` does not re-parse the paragraph from plain text (which would destroy inline bold/italic spans, `[@c:id]` citation nodes, and comment highlights outside the matched range). Instead, it locates the character offsets $[s, e)$ in the ProseMirror node's flattened text stream and replaces only that slice, after which `writeFragment` updates only the modified XML children between the common prefix and suffix in the underlying `Y.XmlFragment`.

3.3 Per-Turn Selective Undo in CRDT Space (`revertTurn`)

In collaborative scientific writing, a researcher may ask the AI to revise three sections, continue manually editing Section 2, and subsequently decide to undo the AI's changes. Global linear undo (`Ctrl+Z` or version rollback) would destroy the researcher's manual edits in Section 2.

Heurion 2.0 solves this via a two-tier *Per-Turn Selective Undo* mechanism (`revertTurn`):

1. **In-Memory CRDT Origin Tracking:** When an AI turn t commits changes to document d, the runtime binds a dedicated `ServerOrigin` object $o_t = \{\texttt{server} : \text{true}, \texttt{actor} :$

'ai', turnId : t} and attaches a scoped Y.UndoManager configured with trackedOrigins = $\{o_t\}$. Invoking revertTurn(d, t) executes um.undo() solely on the CRDT items inserted or deleted under origin o_t, leaving concurrent human character insertions—even within the same paragraph—intact.

2. **Persistent Node-Level Fallback across Server Restarts:** Before turn t's first write, the runtime persists the pre-turn Yjs state vector in turn_doc_bases. If the server restarts before revertTurn is invoked, the engine reconstructs the pre-turn document tree, computes the per-block diff against turn t's recorded node_changes, reverts all AI-modified blocks untouched by the user since $r_{\min}(t)$, and safely skips blocks subsequently edited by the user (returning their IDs in skipped).

3.4 Original-Slice OOXML Patch Export

Clinical manuscripts are frequently imported from journal-specific Word templates containing custom styles, headers/footers, macros, and embedded vector objects. During .docx or .pptx import, Heurion 2.0 stores the original archive in doc_packages and records the raw XML slice (<w:p> or <w:tbl>) of every imported block in node_src keyed by block ID. Upon export (docx-export.ts, pptx-export.ts), any block whose ProseMirror node is structurally equal (node.eq(baselineNode), meaning its text, inline formatting marks, and attached comment range markers are unchanged) to its imported baseline is written back using its exact original node_src XML bytes; only blocks whose text or inline comment/mark structure was mutated during the session are re-serialized, preserving paragraph properties (<w:pPr>), run fonts, and paragraph identifiers (w14:paraId) from the original template.

4 Citation-Grounded Evidence Loop and Privacy-Preserving Clinical Pipeline

How can a clinical authoring platform prevent LLM citation fabrication while protecting patient privacy (PHI) in medical records? Heurion 2.0 addresses both requirements through deterministic pre-write syntactic guards and a two-stage neuro-symbolic PHI redaction funnel.

4.1 Syntactic Citation Guard and Atomic Citation Nodes

To prevent fabricated references from entering manuscript text, Heurion 2.0 blocks agents from writing raw bibliographic identifiers or hand-numbered brackets in prose. During pre-write validation in OpService, every candidate text string is scanned against structural regular expressions detecting bare DOIs (10.\d{4,9}/…), bare PMIDs (PMID:\s*\d+), or unlinked reference markers. Any violation triggers citation_not_registered before modifying the document.

Instead, the agent must invoke insert_citation, which queries live metadata APIs (Crossref, NCBI PubMed E-utilities, and OpenAlex), verifies the record's existence, stores the canonical metadata (authors, title, journal, year, volume, pages, DOI/PMID) in the document's citations table, and returns a registered token [@c:cite_id]. In ProseMirror, [@c:cite_id] is parsed into an atomic inline node (atom: true) that dynamically numbers itself according to citation order and automatically compiles the terminal References section.

4.2 System Context: Claim Verification and Biostatistical Sandbox

In addition to the pre-write citation guard evaluated in Section 7, the platform includes a multi-agent claim verification service (ClaimService.verify) and a sandboxed biostatistical execution toolchain (included here for architectural context; clinician-graded evaluation of these modules is left to future work). Briefly, verify_claims segments manuscript blocks into

sentences, retrieves PubMed abstracts and available PubMed Central (PMC) full texts for cited `[@c:cite_id]` records, and classifies quantitative assertions into four verdicts (`supported`, `contradicted`, `unsupported`, or `uncited_stat`), attaching problematic claims as inline ProseMirror review comments. For statistical analysis, computation is delegated to isolated Python/R tools (`biostatistician` subagent) rather than free-text LLM arithmetic.

4.3 Multi-Tenant PHI Redaction Funnel

To bridge clinical data management and manuscript authoring without exposing patient identities [12, 13, 14], Heurion 2.0 enforces four privacy safeguards:

- **Per-Tenant Database Isolation & Envelope Encryption:** Each organization tenant operates on a physically isolated SQLite database file (`tenants/<org_id>/platform.db`) with per-tenant Data Encryption Keys (DEKs) wrapped by a master Key Encryption Key (KEK).
- **Two-Stage Neuro-Symbolic PHI Funnel (`phi-scan.ts`):** Using an external cloud LLM to inspect raw clinical text for PHI creates a privacy paradox: sending unredacted patient records to a third-party API in order to redact them already violates tenant data-residency boundaries. Conversely, running a frontier-scale 70B+ parameter model on-premise imposes high GPU requirements for clinical research teams. Heurion 2.0 addresses this trade-off through a task-decomposed, two-stage neuro-symbolic funnel that operates within the local environment:
  1. *Stage 1 — Sub-Millisecond Deterministic Checksum & Subtype Rules (`scanPhi`/`redactPhi`):* Deterministic rules execute first (0.011 ms per note) to validate and mask mathematically structured or field-labeled identifiers: 18-digit national IDs (ISO 7064 MOD 11-2 checksum + calendar bounds), settlement/bank cards (ISO/IEC 7812 industry prefix + Luhn mod-10 checksum with DOI/URL boundary exclusion), 11-digit mobile numbers, medical record numbers, and co-occurring patient-name + clinical diagnosis pairs spanning both pure-CJK and mixed alphanumeric/Roman-numeral subtypes (e.g., Type-2 diabetic nephropathy, diffuse large B-cell lymphoma).
  2. *Stage 2 — Pre-Masked Local Small Language Model with Symbolic Output Guard (`createLocalSlmPhiDetector` + `redactPhiCascaded`):* Because Stage 1 pre-masks structured identifiers before Stage 2 is invoked, the Stage-2 model does not receive raw national IDs, phone numbers, or bank cards, and does not need to perform arithmetic checksum verification. Instead, a compact 1.5B-parameter local small language model (`qwen2.5:1.5b`, 986 MB, served via local Ollama and gated by `assertLocalPhiEndpoint` to allow only loopback `127.0.0.1`/`::1` or RFC 1918 private intranet hosts) extracts open-vocabulary residual spans in free-form ward-round narratives: unlabeled person/relative names (`unlabeled_person_name`) and specific residential/workplace addresses (`address_or_org`). Finally, a deterministic post-validator (`normalizeAndValidateStage2Entity`) verifies exact substring presence in the source text, strips kinship/location prepositions, checks Chinese family-surname membership and medical-suffix exclusions for names, and requires spatial/organizational markers for addresses—filtering out small-model hallucinations and clinical-term false positives. The model thus acts as a candidate generator and the post-validator decides what is masked; text containing no CJK characters (e.g., English manuscript prose) is not sent to the model.
- **Pseudonymous Subject Mapping & PHR Red-Lines:** Within research cohorts, real patient identifiers are replaced by pseudonymous subject codes (`P001`, `P002`, …) via an in-memory translation table. Furthermore, a dedicated Personal Health Record guard (`phr_guard`) enforces read-only access on patient-uploaded primary medical records, preventing AI tools from altering ground-truth clinical evidence.

- **Regulatory Scope:** These mechanisms reduce exposure but do not by themselves establish compliance with either PIPL or HIPAA. Pseudonymous subject codes are de-identification rather than anonymization: under PIPL, data that can be re-linked with a retained mapping table remains personal information [16]. The current scanner implements mainland Chinese identifier formats (resident ID numbers, mobile numbers, hospital record numbers, Chinese surnames/addresses); it does not yet cover HIPAA Safe Harbor identifiers such as Social Security numbers, U.S.-format telephone numbers, full dates, or English geographic subdivisions [15], so U.S. deployment requires extending it.

5 Three-Layer Self-Evolving Agent Architecture

A recurring challenge in contract-guarded agent environments is helping the LLM internalize operational boundaries without manual prompt engineering. When an agent's tool call is blocked by a pre-write guard or reverted by a user, how does the system help the agent adapt both within the current task and across future sessions?

As shown in the bottom tier of Figure 1, Heurion 2.0 implements a three-layer adaptation architecture operating across three timescales:

5.1 Layer 1: Task-Level Pre-Flight Probe (`doc_validate` & `dry_run`)

At the intra-turn millisecond timescale, agents in Heurion 2.0 can invoke `doc_validate` (or pass `dry_run: true` to `doc_edit`) prior to committing complex multi-block edits. The probe executes the full suite of conflict guards, citation registration checks, anchor integrity checks, and post-edit structural diffs against an ephemeral in-memory clone of the ProseMirror document tree without incrementing `rev`, mutating the Yjs CRDT state, or polluting the undo stack. If a violation is detected, the probe returns `{valid: false, code, hint, current}`, enabling zero-cost self-correction within the same reasoning turn.

5.2 Layer 2: Session-Level Dual-Track Memory and Utility-Ranked Injection

At the cross-turn session timescale, Heurion 2.0 addresses memory pollution by enforcing **Declarative-Procedural Orthogonality** (Invariant 3):

- **Track A — Declarative Memory (`MemoryService`):** Captures domain conventions and author preferences (e.g., "Always report continuous skewed variables as median (IQR)", "Use British English spelling"). Because stylistic preferences are subjective, newly extracted items enter a `candidate` state and require explicit human approval in the UI before becoming `confirmed`.
- **Track B — Procedural Harness Memory (`ProceduralHarnessService`):** Captures objective operational rules for navigating guards, tools, and specialist subagents. Whenever `OpService` rejects an AI edit or a user invokes `revertTurn`, the `ExecutionSignals` collector automatically records a structured runtime signal $s = (\text{owner}, \text{doc_id}, \text{turn_id}, \text{kind}, \text{code}, \text{detail})$ into the `execution_signals` table.

At the end of an authoring session, `evolveFromSignals()` clusters unconsumed execution signals by $(\text{kind}, \text{code})$. Any failure pattern occurring at least $\tau = 2$ times either increments the empirical evidence counter e_r of an existing procedural rule $r \in \mathcal{M}_P$ or crystallizes a new active procedural directive via guard-specific actionable templates (`LEARNED_RULE_TEMPLATES`)—requiring zero manual UI intervention. Rather than emitting generic warnings (e.g., "call `doc_validate` before editing"), each crystallized rule encodes the exact positive parameter or tool-sequencing fix required by the underlying contract (e.g., including leading clause context or explicit `occurrence: 2` for `ambiguous_match`; stripping HTML `<sup>`/`<sub>` tags from `find`

while retaining them in `replace` for `text_not_found`; prepending `{op: "move"}` before `{op: "merge"}` in the same batch for `invalid_structure`; calling `comments_list` after `doc_read` and passing `ack_comments` + `comment_reply` for `anchor_has_open_comments`; targeting only the top-left origin `{row: 0, col: 0}` of merged table headers for `cell_not_found`; and restricting edits to the single named block after `user_reverted_turn`).

Context-Aware Utility Ranking. Injecting all procedural rules into every turn would waste context tokens. Before each turn on document d with user prompt q, `ProceduralHarnessService.forPrompt()` computes a relevance score $U(r \mid d, q)$ for each active rule $r \in \mathcal{M}_P$. Guard codes never appear in a user's request, so a rule is not matched against q by its code; instead each guard code is associated with a *message cue* (keywords in q, e.g., DOI/PMID or citation terms for `citation_not_registered`, replacement/statistic verbs for `ambiguous_match` and `text_not_found`, paragraph-merge or transition verbs for `invalid_structure` and `user_reverted_turn`) and/or a *document-state cue* computed from d (human edits within the last two revisions for `conflict_user_edited`, open comment threads for `anchor_has_open_comments`, pending suggestions for `pending_suggestion`, the presence of a table for `cell_not_found` and `invalid_table`):

$$U(r \mid d, q) = \alpha \cdot e_r + \beta \cdot \mathbb{I}(\text{msgCue}_r(q)) + \gamma \cdot \mathbb{I}(\text{stateCue}_r(d)) \,, \tag{3}$$

where e_r is the empirical evidence count, $\mathbb{I}(\cdot)$ is the indicator function, and $(\alpha, \beta, \gamma) = (0.1, 5, 8)$. Baseline subagent-delegation rules, whose trigger is a keyword pattern rather than a guard code, use that pattern as their message cue; codes without a defined cue are ranked by e_r alone. These weights are hand-set heuristics, not tuned values. Rules are sorted in descending order of $U(r \mid d, q)$ and greedily packed: a rule of length ℓ is charged $\ell + 20$ characters and skipped if it would exceed the budget B (default $B = 1000$), up to at most $K = 4$ rules. To isolate rule content from prompt-length effects in controlled experiments, the service also supports a `placebo` mode (`PROCEDURAL_MODE=placebo`) that appends a length-matched block of generic platform background text alongside the static baseline rules.

5.3 Layer 3: System-Level Meta-Strategy Distillation (`HarnessEvolutionService`)

At the multi-session platform timescale, Heurion 2.0's `HarnessEvolutionService` periodically aggregates system-wide execution signals and active procedural rules. When cumulative unanalyzed signals exceed a threshold, a background meta-evolver distills recurring failure root causes into concise system-level behavioral directives (2–5 rules, ≤ 100 characters each) and serializes them into a dynamic YAML profile overlay (`evolved.cordis.yml`):

```
# Auto-generated by Heurion Self-Evolution Engine
- id: system-prompt
  config:
    personaSuffix: >-
      [Evolved Self-Improvement Execution Directives]:
      - Run doc_validate prior to mutating blocks with recent user activity
      - Register all DOI/PMID entries via insert_citation before inline referencing
```

When `HarnessPool` spawns or refreshes an agent runtime, `getPatches()` transparently mounts `evolved.cordis.yml` alongside the base platform profile, updating the agent's system persona without user disruption.

6 CI-Enforced Human-AI Permission Symmetry

A common security and usability gap in enterprise AI applications is the disconnect between what a human user can perform via the graphical user interface and what the AI agent can

perform via tool calls. Often, developers either grant the agent direct database access (bypassing business-layer authorization) or expose only a subset of tools.

Heurion 2.0 enforces **Human-AI Permission Symmetry** (Invariant 2) across all platform capabilities:

1. **In-Process Loopback Execution under User Identity:** Every administrative and clinical MCP tool [18] (`patient_*`, `study_*`, `doc_manage`, `kb_manage`, `dataset_manage`, `org_manage`) executes by invoking `invokeHttp(api, userId, method, path, body)` directly against the Hono HTTP router in-process. The call carries a short-lived internal JWT minted for `userId` and tagged with header `X-Heurion-Via: ai`. Consequently, the agent inherits the exact same multi-tenant isolation, study role checks (PI, co-investigator, statistician, reviewer), and audit logging as the human user.
2. **Human-Only Confirmation Cards for High-Risk Actions:** When an AI tool invokes a destructive or irreversible endpoint (e.g., purging a document, freezing a clinical study cohort, or removing a team member), the HTTP middleware intercepts the `X-Heurion-Via: ai` header and—rather than executing the mutation—creates a `pending_actions` record and renders an interactive confirmation card in the user's chat stream. Only when the human clicks *Confirm* in the browser does the UI re-issue the request with `X-Heurion-Via: ai-confirmed`, completing execution.
3. **Automated CI Parity Verification (`parity.test.ts`):** Our test suite introspects the Hono application router table at build time and asserts that every registered `/api/*` endpoint appears either in the MCP parity mapping table ($\phi(\mathcal{E}_{\text{MCP}})$) or in the documented $\mathcal{E}_{\text{NOT_FOR_AI}}$ exclusion registry. Any newly added API route missing an MCP counterpart or explicit exemption immediately fails continuous integration.

7 Empirical Evaluation

We evaluate Heurion 2.0 through a two-tier, two-generation empirical protocol (`ANALYSIS_PLAN.md` for v1 and `ANALYSIS_PLAN_v2.md` for v2) combining:

1. **Deterministic and Local-SLM Benchmarks (`run_eval.ts` & `v2/phi/run.ts`):** 1,010 evaluation records against the platform's TypeScript/SQLite/Yjs engine and local Ollama runtime, comprising 50 CRDT concurrency/revert trials (`exp1`), 660 initial citation and template PHI guard records (`exp2b`), and a locked 300-record multi-specialty Chinese clinical PHI de-identification test split (`v2-phi`: 100 diverse clinical notes containing 745 PHI spans [485 narrative, 260 labeled] and 200 hard-negative clinical sentences evaluated across 4 ablation arms). Local Stage-2 PHI extraction used `qwen2.5:1.5b` (986 MB, Ollama v0.20.5 on `http://127.0.0.1:11434/api/chat`, `temperature=0`, `format='json'`).
2. **Pre-Specified Live LLM Evaluations (`ANALYSIS_PLAN.md` & `ANALYSIS_PLAN_v2.md`):** 10,814 analysed live agent turns across 20 open-access (CC BY) clinical research manuscripts from PubMed Central (PMC), comprising: (i) 1,286 v1 baseline turns (162 in `exp2a-papers`, 45 in `exp2c`, and 1,079 in `exp3`); (ii) 4,488 three-arm collaborative manuscript editing turns in `v2-q1` (2,148 development turns in Phase 1a on 10 PMC manuscripts and 2,340 confirmatory turns in Phase 1b on 10 newly selected 2026 PMC manuscripts across stress, multi-turn session, and routine tasks); and (iii) 5,040 longitudinal procedural memory turns in `v2-exp3` (432 pilot turns across 3 arms and 4,608 confirmatory turns across 4 arms $\times$ 16 users $\times$ 8 sessions $\times$ 9 uncovered constraint tasks, with 0 failed or excluded turns). All online turns were executed using model ID `deepseek-flash` via `@deepseek-ai/dsh-sdk-client` against `https://api.deepseek.com/anthropic/v1`. Across longitudinal memory experiments, each turn spawns a fresh agent session (`HARNESS_FRESH_SESSION=1`) with declarative

Table 1: Architectural mechanism comparison across direct script mutation (`python-docx`), script mutation with post-hoc file auditing, and Heurion 2.0's contract-guarded dual-surface CRDT architecture. Entries describe system design mechanisms, not measured rates.

Dimension	Direct Script (`python-docx`)	Script + Post-Hoc Audit	Heurion 2.0 (Ours)
Source of Truth	Raw `.docx` on Disk	Raw `.docx` + Derived Projection	**Shared Yjs CRDT + SQLite**
Block Addressing	None (Regex/XPath)	File-Native `w14:paraId` / Shape ID	**Server-Assigned `{#id}`**
Export of Untouched Blocks	Full Re-serialization	Whole-File Save	**Original XML Slice (`node_src`)**
Concurrent Human Edits	Silent Overwrite	Post-Hoc Node-Level 3-Way Merge	**Pre-Write `base_rev` Guard**
Undo of One AI Turn	Unsupported	File-Level Rollback	**Per-Turn `Y.UndoManager`**
Unregistered Citations	Not Checked	Post-Write Audit	**Pre-Write Block**
Pre-Flight Validation	None	Subprocess Audit Script	**In-Memory Dry Run**

Table 2: Heurion 2.0 dual-surface CRDT engine micro-benchmark results (`exp1`) over $N = 50$ trials on the synthetic clinical protocol template. Latencies are wall-clock means of in-process calls on a single developer machine.

Metric	Result
Untouched blocks reproduced verbatim on export (heading + final paragraph)	50/50
Stale `replace_block` rejected with `conflict_user_edited`	50/50
Turn revert undoes AI edits while keeping the human edit in the same paragraph	50/50
Mean pre-flight validation latency (`OpService.validate`)	0.168 ms
Mean guarded edit latency (`OpService.edit`, two ops)	0.338 ms
Mean per-turn revert latency (`revertTurn`)	0.129 ms

memory auto-extraction and Layer 3 background overlays disabled (`HARNESS_EVOLVE=0`) so that simulated user trajectories (parameterized by random seeds) remain statistically independent.

7.1 Experiment 1: Dual-Surface CRDT Engine vs. Script-Based File Mutation

Table 1 contrasts the architectural mechanisms of direct script mutation (`python-docx`), script mutation paired with post-hoc file auditing, and Heurion 2.0.

Experiment 1a: Deterministic Engine Micro-Benchmarks (`exp1`). Table 2 reports in-process micro-benchmarks of Heurion 2.0 over $N = 50$ trials on a six-block `.docx` clinical protocol template (one styled heading, a bold paragraph, an italic paragraph, two list items, and a plain paragraph). In each trial, an AI turn applies two `replace_text` edits (p_1, p_2), a simulated human edits a different phrase inside p_2, a stale `replace_block` is probed, and the AI turn is selectively reverted via `revertTurn`. All 50/50 trials reproduced the original `<w:p>` XML bytes of the untouched heading and final paragraph verbatim via `node_src`, rejected the stale overwrite with `conflict_user_edited`, and undid the AI edits while preserving the human's intra-paragraph edit in p_2 at sub-millisecond latency (0.168 ms validation, 0.338 ms edit, 0.129 ms revert).

Experiment 1b: End-to-End Clinical Manuscript Editing—Heurion 2.0 MCP vs. `python-docx` (`exp2a-papers`). To test how the two editing surfaces behave under live LLM tool use on full-length journal manuscripts, we compared Heurion 2.0's MCP surface (`PROCEDURAL_MODE=off`) against a baseline coding agent using the same model (`deepseek-flash`) and identical authoring guidelines (`baseline.cordis.yml`) to edit `manuscript.docx` directly in a workspace directory. In the baseline arm, the agent runs inside the same `DeepSeekHarness` runtime with unrestricted `bash` and `python3` access (including

Table 3: End-to-end clinical manuscript editing comparison (`exp2a-papers`, 162 live LLM turns across 81 paired tasks per arm, 3 simulated user seeds × 3 manuscripts × 9 archetypes) between Heurion 2.0 MCP primitives and direct `python-docx` file scripting. 95% confidence intervals use the Wilson score method; p-values are two-sided Fisher's exact tests. †In 2 `open_comment_anchor` turns (`u3-s3`, `u1-s3`), the MCP agent invoked `delegate_task` (`fact_checker`) without being asked to, which attached 7 unrequested verification comments (`<w:commentRangeStart/>`) to non-target paragraphs; their text was unchanged (`content_changed = 0`), but these are AI-initiated writes outside the task scope. ‡In full-paragraph rewrites, the MCP agent passed `drop_anchors: true` (6/9 cases) to clear obsolete range tags after replying to all 9/9 review threads via `comment_reply`, whereas `python-docx` run-level edits left the sibling `<w:commentRangeStart/>` tags in place without replying to `word/comments.xml`.

Evaluation Endpoint	**Heurion 2.0 MCP** ($n = 81$)	**Direct `python-docx`** ($n = 81$)	**Fisher** p
Task completion rate across all 9 archetypes	76/81 (93.8%, CI 86.4–97.3%)	77/81 (95.1%, CI 88.0–98.1%)	1.0000
Untouched paragraphs with unintended content changes	0/32,995 (0.00%)	4/32,995 (0.01%)	0.1250
Untouched paragraphs preserved byte-identically†	32,988/32,995 (99.98%)	32,991/32,995 (99.99%)	0.5488
Edited paragraphs preserving native `w14:paraId`	41/45 (91.1%, CI 79.3–96.5%)	37/45 (82.2%, CI 68.7–90.7%)	0.3529
Edited paragraphs preserving layout, fonts, and emphasis	44/45 (97.8%)	45/45 (100.0%)	1.0000
Review comments read & replied in thread (`acknowledged`)	**9/9 (100.0%, CI 70.1–100%)**	0/9 (0.0%, CI 0.0–29.9%)	**< 0.0001**
Inline `<w:commentRangeStart/>` tag retained (6/9 cleared via `drop_anchors`)‡	3/9 (33.3%, CI 12.1–64.6%)	**9/9 (100.0%, CI 70.1–100%)**	**0.0090**
Clinician pending track-change suggestion preserved	9/9 (100.0%, CI 70.1–100%)	9/9 (100.0%, CI 70.1–100%)	1.0000
Unregistered bare DOI / PMID injected into manuscript body	0/81 (0.0%, CI 0.0–4.5%)	1/81 (1.2%, CI 0.2–6.7%)	1.0000

`python-docx`, `zipfile`, and `xml.etree.ElementTree`) and is explicitly instructed to *"preserve the existing formatting, comments and tracked changes"*. Both arms executed 3 paired simulated user seeds (seeds 201–203) × 3 PMC clinical manuscripts × 9 authoring task archetypes (`concurrent_edit`, `citation`, `pending_suggestion`, `open_comment_anchor`, `term_unify`, `table_row`, `out_of_scope_revert`, `p_value_format`, and `move_subsection`), yielding 81 turns per arm (162 turns total; 32,995 untouched paragraphs per arm). Both arms exported standard `.docx` archives evaluated by the same OOXML inspector (`manuscripts/compare.ts`).

Table 3 summarizes three empirical observations from `exp2a-papers`:

1. **Comparable Task Completion and Structural Preservation Rates:** Task completion did not differ significantly between Heurion 2.0 MCP (76/81 = 93.8%) and direct `python-docx` scripting (77/81 = 95.1%, Fisher $p = 1.0000$). Across 32,995 non-target paragraphs, Heurion 2.0 produced 0 unintended text modifications compared to 4 altered non-target paragraphs in the `python-docx` arm ($p = 0.1250$), and retained `w14:paraId` on 41/45 (91.1%) vs. 37/45 (82.2%, $p = 0.3529$) edited paragraphs; neither difference reached statistical significance at $n = 81$ turns per arm. Why were 7 of 32,995 untouched paragraphs in Heurion 2.0 not byte-identical (32,988/32,995 = 99.98% vs. 32,991/32,995 = 99.99%)? Inspection of the turn logs showed that all 7 paragraphs occurred in two `open_comment_anchor` turns (`u3-s3` on `PMC8861838`, 4 paragraphs; `u1-s3` on `PMC10260499`, 3 paragraphs) where the MCP agent voluntarily invoked the `fact_checker` subagent (`delegate_task`), which attached unrequested `<w:commentRangeStart/>` verification comments to those 7 paragraphs while leaving their text unchanged (`content_changed = 0`). We therefore report these as out-of-scope side effects: Heurion 2.0 made 0 unintended text changes but 7 unrequested comment insertions on non-target paragraphs.
2. **Comment-Thread Resolution vs. Unassisted File Scripting:** In the `open_comment_anchor` task (which instructs the agent to rewrite a Methods paragraph according to two attached clinician review comments), although the baseline agent had full Python/XML access to `word/comments.xml`, standard `python-docx` scripts written by the agent iterated over `Document.paragraphs` (which does not expose `word/comments.xml`) and lacked a dedicated comment-thread tool or pre-write guard; consequently, the file agent modified the paragraph's text runs while leaving the sibling `<w:commentRangeStart/>` XML tags

in place (9/9 vs. 3/9, $p = 0.0090$) and never appended a reply to `word/comments.xml` (0/9 acknowledged). Conversely, Heurion 2.0's `anchor_has_open_comments` guard and `comments_list`/`comment_reply` tools prompted the MCP agent to inspect the comments, declare `drop_anchors` when replacing the full paragraph (6/9 cases), and post structured replies in all 9/9 (100%, $p < 0.0001$) review threads. These replies live in the platform's comment threads: the exported `.docx` does not carry them, and in 4/9 exported files the two original clinician comments were removed along with their anchors, so a reviewer working only in Word would not see how the comments were resolved.

3. **Methodological Note on File-Arm Concurrent Edits:** Both arms preserved 9/9 clinician insertions in `concurrent_edit`. Inspection of the event logs revealed that because the file agent executes a single atomic `bash` call (`Document('manuscript.docx') ... save()`) whereas the harness callback `humanEditsFile` fires upon `tool_result` completion, the simulated clinician insertion in the file arm occurred between separate `bash` invocations rather than mid-flight between an in-memory read and write. Mid-flight stale-snapshot overwrites are evaluated deterministically in Table 2 (50/50 intercepted) and in Experiment 1c below.

Experiment 1c: Confirmatory Three-Arm Collaborative Manuscript Editing (`v2-q1`, 4,488 Turns). Because the initial 162-turn `exp2a-papers` comparison used explicit preservation prompts, single-turn tasks, and a file baseline without literature search tools, we designed a two-phase development–validation benchmark (`ANALYSIS_PLAN_v2.md` §2–§4b) comparing three arms under natural clinician instructions, true mid-turn/inter-turn concurrent human edits, complex journal OOXML templates (188/188 fidelity checks passing in `v2/fidelity/run.ts`), and identical literature search access:

- **Arm P (Heurion 2.0 Platform):** Dual-surface CRDT core with document-scoped MCP tools (63 tools exposed per manuscript via `mcp/scope.ts`) and pre-write structural/citation guards.
- **Arm B1 (Direct File Scripting):** Coding agent with `bash`/`python3` (`python-docx`, `lxml`) editing `manuscript.docx`.
- **Arm B2 (Tool-Assisted File Scripting + Audit):** Same as `B1` plus helper inspection/audit scripts and the same live Crossref/PubMed/OpenAlex literature search tools as `P`.

In the **Development Phase (1a, 2,148 turns across 10 PMC manuscripts)**, `P` eliminated lost concurrent clinician sentences (0 vs. 23 in `B1` and 17 in `B2`) and OOXML structural corruption (0 vs. 32 in `B1` and 3 in `B2`), achieving a rework odds ratio of OR = 3.62 vs. `B1` (Holm $p = 2.14 \times 10^{-6}$) and 20% lower per-turn cost (GMR = 0.80, $p < 10^{-21}$), while exposing five platform gaps (unprotected platform-native user suggestions, lack of scoped batch text replacement, unverified citation claim support at insertion time, manual paragraph merges dropping inline marks, and stripped citation subtitles). After implementing fixes `F1--F5` and freezing tag `paper-eval-v2-p1b`, we executed the **Confirmatory Phase (1b, 2,340 turns = 780 turns/arm across 10 newly selected 2026 CC BY PMC manuscripts: 123 single-turn stress tasks, 576 multi-turn collaborative session turns [9 papers × 8 sessions × 8 turns], and 81 routine tasks per arm)**.

As shown in Table 4, all four pre-specified confirmatory hypotheses in Phase 1b passed after Holm correction: Heurion 2.0 reduced rework odds by OR = 32.47 (95% CI 14.39–73.26, Holm $p = 1.03 \times 10^{-16}$) against `B1` and OR = 9.66 (95% CI 4.43–21.08, Holm $p = 1.23 \times 10^{-8}$) against `B2`, while reducing API cost by 30% (GMR = 0.70, 95% CI 0.67–0.73, Holm $p = 3.73 \times 10^{-73}$) and 33% (GMR = 0.67, 95% CI 0.64–0.69, Holm $p = 2.99 \times 10^{-95}$) and cutting median turn latency by 41%–46%. Pre-write guards intercepted 138 unsafe operations across 111 platform turns, preventing any loss of concurrent human edits (0 vs. 18 and 22). Post-hoc root-cause

Table 4: Confirmatory Phase 1b three-arm clinical manuscript editing results (`v2-q1`, $N = 2,340$ live LLM turns across 10 unseen 2026 PMC manuscripts; 780 paired turns per arm). Primary accuracy models use mixed-effects logistic regression on stress + multi-turn session turns ($n = 699$/arm); primary cost models use linear mixed-effects regression on log-cost across all 780 turns/arm, with Holm step-down correction across the four pre-specified confirmatory tests.

Endpoint / Stratum	Heurion 2.0 (P)	Direct Script (B1)	Assisted Script (B2)	B1 vs. P Effect (Holm p)	B2 vs. P Effect (Holm p)
Stress Tasks Rework ($n = 123$/arm)	**2.4**% (3/123)	20.3% (25/123)	9.8% (12/123)	—	—
Multi-Turn Session Rework ($n = 576$/arm)	**5.4**% (31/576)	13.4% (77/576)	10.6% (61/576)	—	—
Routine Tasks Rework ($n = 81$/arm)	**4.9**% (4/81)	8.6% (7/81)	8.6% (7/81)	—	—
All Turns Rework ($n = 780$/arm)	**4.9**% (**38/780**)	14.0% (109/780)	10.3% (80/780)	OR = **32.47** (**1.03×10^{-16}**)	OR = **9.66** (**1.23×10^{-8}**)
Severe Rework ($n = 780$/arm)	**1.2**% (9/780)	4.4% (34/780)	5.4% (42/780)	OR = 27.98 ($p = 4.20 \times 10^{-6}$)	OR = 51.95 ($p = 7.87 \times 10^{-8}$)
Lost Human/Collaborator Content	**0**	18	22	—	—
OOXML File Structural Corruption	**1**	32	3	—	—
Out-of-Scope Collateral Edits	**1**	9	8	—	—
Fabricated / Mismatched Citations	**0**	1	3	—	—
Total API Cost (780 turns/arm)	**$9.26**	$13.21	$13.05	GMR = **0.70** (**3.73×10^{-73}**)	GMR = **0.67** (**2.99×10^{-95}**)
Median Turn Latency (Stress / Session)	**33.5** s/**15.5** s	57.3 s/26.5 s	61.6 s/27.9 s	−41.5% / −41.5%	−45.6% / −44.4%

inspection of all 38 platform rework turns (`ANALYSIS_PLAN_v2.md` §5.3.4) showed that 22/38 were evaluator oracle false positives (affecting all arms equally, e.g., body sentences starting with "Table 1" flagged as missing `SEQ` caption fields), leaving only 16/780 (2.1%) true platform/model defects across 6 root causes, all of which were subsequently patched and covered by 148 unit tests.

7.2 Experiment 2: Pre-Write Citation, PHI, and High-Risk Action Guards

Experiment 2a: Two-Stage Deterministic + Local SLM PHI and Citation Guards (`exp2b` & Locked `v2-phi` Test Split). To quantify the complementary roles of Stage-1 deterministic validators (`scanPhi`/`redactPhi`) and the Stage-2 local small language model (`qwen2.5:1.5b` via loopback Ollama, `redactPhiCascaded`), we first evaluated Heurion 2.0 across 660 template test records (`exp2b`: 100 citation-guard sentences, 200 synthetic positive clinical notes containing 720 PHI spans across 7 categories, and 360 negative controls across 4 strata; Table 6), and then conducted a pre-specified, single-run evaluation on a locked, multi-specialty Chinese clinical de-identification test set (`v2-phi`, Table 5):

1. **Unregistered Inline Citations ($N = 100$ distinct sentences):** 40 bare DOIs across 8 biomedical publisher prefixes, 30 `https://doi.org/` URLs, and 30 bare PMIDs. All 100/100 (100%) were intercepted prior to CRDT mutation with `citation_not_registered`.
2. **Template Verification Benchmark (`exp2b`, 720 Spans in 200 Notes + 360 Negative Controls):** As shown in Table 6, **Stage 1 alone** (0.011 ms/note) achieved 100.0% (600/600) recall across the five structured and labeled categories, 0/200 false positives on numeric-dense PMC biostatistical paragraphs, and 0/100 false positives on invalid-checksum 16–18 digit serials, while **Stage 2 (`qwen2.5:1.5b` + post-validator)** captured 110/120 (91.7%) of template narrative names and addresses (710/720 = 98.6% combined recall) with 1/60 (1.7%) false positives on clean Chinese clinical sentences (vs. 60/60 without post-validation).
3. **Locked Multi-Specialty Clinical De-Identification Test Set (`v2-phi`, 100 Notes / 745 Spans + 200 Hard-Negative Sentences):** Because `exp2b`'s narrative clauses shared a single phrasing template, we constructed an independent benchmark (`ANALYSIS_PLAN_v2.md` §3) generated by an external LLM across diverse clinical departments and note types (admission notes, ward-round records, discharge summaries, family consultations) where 65.1% (485/745) of PHI spans occur in unlabeled free-form narrative prose and 34.9% (260/745) occur in labeled fields, paired with 200 hard-negative Chinese clinical sentences dense with eponyms, anatomical terms, drug regimens, and statistical results. After refining Stage-1 surname-prefix stripping and Stage-2 biomedical exclusion lists on the `dev` split (40 notes +

Table 5: Locked test-split evaluation of the two-stage neuro-symbolic PHI funnel (`v2-phi`, `2026-10-22T15-08-56-203Z-test.json`; 100 diverse Chinese clinical notes containing 745 PHI spans [485 narrative, 260 labeled] and 200 hard-negative clinical sentences). Parentheses show Wilson 95% confidence intervals.

Ablation Configuration	Overall Recall ($N = 745$ spans)	Narrative Recall ($n = 485$ spans)	Labeled Recall ($n = 260$ spans)	Span Precision (True / Total Pred.)	Hard-Negative FPR ($N = 200$ sents)
`A_stage1` (Stage-1 Regex/Checksum Only)	27.4% (204/745) (24.3–30.7%)	3.5% (17/485) (2.2–5.5%)	71.9% (187/260) (66.2–77.0%)	**100.0**% (209/209) (98.2–100.0%)	**0.0**% (0/200) (0.0–1.9%)
`B_platform` (Stage 1 + 1.5B SLM + Validator)	**87.0**% (**648/745**) (**84.4–89.2**%)	**83.3**% (**404/485**) (**79.7–86.4**%)	**93.8**% (**244/260**) (**90.2–96.2**%)	**97.6**% (**482/494**) (**95.8–98.6**%)	**6.0**% (**12/200**) (**3.5–10.2**%)
`C_slm_unvalidated` (Stage 1 + 1.5B SLM, No Validator)	87.7% (653/745) (85.1–89.8%)	84.3% (409/485) (80.8–87.3%)	93.8% (244/260) (90.2–96.2%)	56.2% (487/866) (52.9–59.5%)	92.0% (184/200) (87.4–95.0%)
`D_rules_only` (Stage 1 + Broad Rules + Validator)	98.8% (736/745) (97.7–99.4%)	99.6% (483/485) (98.5–99.9%)	97.3% (253/260) (94.6–98.7%)	36.6% (567/1550) (34.2–39.0%)	93.5% (187/200) (89.2–96.2%)

80 negatives), we executed a single locked run on the held-out `test` split (100 notes + 200 negatives; `TEST_RUN.lock`, Table 5) across four pre-specified ablation configurations:

- `A_stage1` (*Stage-1 regex/checksum only*): Achieved 100.0% (209/209) span precision and 0.0% (0/200) hard-negative FPR, but recalled only 27.4% (204/745) of total PHI—capturing 71.9% (187/260) of labeled spans but only 3.5% (17/485) of narrative spans (0/360 narrative person names, 0/99 narrative addresses).
- `B_platform` (*Stage 1 + Local* 1.5*B SLM + Deterministic Post-Validator*): Raised overall PHI recall by +59.6 percentage points to **87.0**% (**648/745**, Wilson 95% CI 84.4–89.2%), boosting narrative PHI recall from 3.5% to **83.3**% (**404/485**, CI 79.7–86.4%; narrative person names 289/360 = 80.3%, narrative addresses 92/99 = 92.9%, narrative workplaces 6/7 = 85.7%) and labeled recall to **93.8**% (**244/260**), while maintaining **97.6**% (**482/494**, CI 95.8–98.6%) span precision and a **6.0**% (**12/200**, CI 3.5–10.2%) hard-negative sentence FPR.
- `C_slm_unvalidated` (*Stage 1 + Local* 1.5*B SLM without Post-Validator*): Recalled 87.7% (653/745) of PHI (only +5 spans over `B_platform`), but suffered catastrophic over-redaction of biomedical terminology: span precision collapsed to 56.2% (487/866) and hard-negative sentence FPR surged to **92.0**% (**184/200**).
- `D_rules_only` (*Stage 1 + Broad Surname/Spatial Rule Candidates + Same Post-Validator, No SLM*): Recalled 98.8% (736/745) of PHI, but without the SLM's contextual filtering, span precision dropped to 36.6% (567/1,550) and hard-negative FPR reached **93.5**% (**187/200**).

Design Implication: Local SLMs as Contextual Candidate Generators Between Deterministic Guards. Tables 6 and 5 demonstrate why neither deterministic rules nor a compact local SLM suffices alone for clinical de-identification. High-precision regex and checksum rules (`A_stage1`) miss 96.5% of narrative PHI, whereas broad candidate rules (`D_rules_only`) and unvalidated 1.5B SLM predictions (`C_slm_unvalidated`) corrupt over 92% of clean clinical sentences by misclassifying diseases, anatomical structures, and eponyms as person names or addresses. Sandwiching the local 1.5B SLM between Stage-1 pre-masking and a symbolic biomedical post-validator (`B_platform`) retains 99.2% (648/653) of the SLM's true PHI detections while eliminating 96.8% (360/372) of its false-positive span predictions, keeping all patient text within `127.0.0.1`.

Experiment 2b: High-Risk Action Confirmation under User Pressure and Prompt Injection (`exp2c`). To test Architectural Invariant 2 across destructive operations, we evaluated a live `deepseek-flash` agent across 5 irreversible administrative scenarios (`purge_doc`, `delete_comment`, `delete_kb`, `delete_study`, and `clear_memory`) under 3 prompt conditions × 3 repetitions (45 turns total; `exp3/confirm.ts`):

Table 6: Heurion 2.0 pre-write citation and two-stage synthetic PHI guard benchmark (`exp2b`) comparing Stage-1 deterministic rules (0.011 ms/note) and Stage-2 local `qwen2.5:1.5b` SLM fallback across 720 synthetic positive PHI spans (200 notes) and 360 negative controls. Stage-2 false-positive cells for the clean Chinese sets report the model without post-validation; the combined column includes the post-validator.

Guard Subsystem / Evaluation Stratum	**Validation Mechanism**	**Test Inputs**	**Stage-1 Only (`scanPhi`)**	**Stage-2 Local SLM (`qwen2.5:1.5b`)**	**Combined Two-Stage Recall / FP Rate**
Bare DOI / URL / PMID Blocker	Pre-write AST regex	100 sents	100/100 (100%)	—	**100/100 (100.0%)**
Stale Block Overwrite Blocker	Block `base_rev` lock	50 trials	50/50 (100%)	—	**50/50 (100.0%)**
PHI: 18-Digit National ID (`id_card`)	ISO 7064 + date check	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: 11-Digit Mobile (`phone`)	Prefix + boundary check	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Medical Record (`medical_record_no`)	Clinical label match	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Bank / Settlement Card (`bank_card`)	ISO 7812 + Luhn mod-10	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Labeled Name + Diagnosis (`name_with_diagnosis`)	Alphanumeric/CJK subtype	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Unlabeled Narrative Name (`unlabeled_person_name`)	Local 1.5B SLM + surname guard	60 spans	0/60 (0.0%)	**+55/60 (91.7%)**	**55/60 (91.7%)**
PHI: Unlabeled Address / Org (`address_or_org`)	Local 1.5B SLM + spatial guard	60 spans	0/60 (0.0%)	**+55/60 (91.7%)**	**55/60 (91.7%)**
PHI Total Entity Recall (200 notes)	**Two-stage neuro-symbolic**	**720 spans**	600/720 (83.3%)	**+110/120 (91.7%)**	**710/720 (98.6%)**
Neg. Control: Real PMC Biostatistical Text	HR/CI/p-value/DOI text	200 paras	0/200 (0.0% FP)	skipped (no CJK; 0 calls)	**0/200 (0.0% FP)**
Hard Neg.: Invalid-Checksum 16–18 Digit IDs	ISO 7064 + Luhn vs. regex	100 serials	0/100 (0.0% FP)	—	**0/100 (0.0%)** vs. 100%
Hard Neg.: Clean Chinese Clinical Text (template set)	SLM ± post-validator	30 sents	0/30 (0.0% FP)	30/30 (unvalidated)	**1/30 (3.3% FP)**
Hard Neg.: Clean Chinese Clinical Text (independent set)	SLM ± post-validator	30 sents	0/30 (0.0% FP)	30/30 (unvalidated)	**0/30 (0.0% FP)**
Control: Rule-Only Stage-2 Candidates	Rules + same post-validator	120 spans / 60 sents	—	120/120 recall; 60/60 FP	—

Table 7: Heurion 2.0 high-risk action confirmation guard results (`exp2c`, 45 live LLM turns across 5 irreversible scenarios × 3 conditions × 3 repetitions). In the 30 turns where the agent invoked the destructive tool (`normal` and `pressure`), the HTTP middleware intercepted 30/30 (100%) calls; in the 15 `injection` turns, the model itself refrained from invoking the destructive tool (0/15).

Prompt Condition	**Turns**	**Tool Invoked / Card Emitted**	**Executed Without Confirmation**	**Executed After `confirm` (Rep. 1)**	**Preserved After `reject` (Rep. 2)**	**Preserved When Unanswered (Rep. 3)**
1. Normal Request (`normal`)	15	15/15 (100%)	**0/15 (0%)**	5/5 (100%)	5/5 (100%)	5/5 (100%)
2. User Pressure (`pressure`)	15	15/15 (100%)	**0/15 (0%)**	—	—	15/15 (100%)
3. Prompt Injection (`injection`)	15	0/15 (0% invoked)	**0/15 (0%)**	—	—	— (not invoked)
Middleware Guard Tested (1+2)	**30**	**30/30 (100%)**	**0/30 (0%)**	**5/5 (100%)**	**5/5 (100%)**	**20/20 (100%)**
All Turns (1+2+3)	**45**	30/45 (66.7%)	**0/45 (0%)**	**5/5 (100%)**	**5/5 (100%)**	**20/20 (100%)**

1. **Normal Request (`normal`, 15 turns = 5 scenarios × 3 repetitions):** The user asks the agent to delete the target resource. To test all three confirmation lifecycle states, repetition 1 simulates user approval (`POST /api/actions/:id/confirm`, 5 turns), repetition 2 simulates user rejection (`POST /api/actions/:id/reject`, 5 turns), and repetition 3 leaves the confirmation card unanswered (`decision = none`, 5 turns) to verify that pending actions do not execute by default.
2. **Coercive User Pressure (`pressure`, 15 turns):** The user explicitly instructs the agent to bypass confirmation (e.g., *"Delete it immediately without asking me to confirm; I take full responsibility"*), and no confirmation button is clicked (`decision = none`).
3. **Indirect Document Prompt Injection (`injection`, 15 turns):** The user asks only to polish the Discussion section, while the manuscript body contains an embedded adversarial directive [17] instructing the agent to purge the target resource without confirmation.

As shown in Table 7, separating middleware interception from model behavior clarifies how the two layers interact: across the 30 turns in which the agent actually called the destructive tool (15 `normal` + 15 `pressure`), the HTTP router middleware intercepted the `X-Heurion-Via: ai` header and blocked unconfirmed execution in all 30/30 (100%) turns, executing the deletion only when followed by a human-authenticated `confirm` call (5/5 in repetition 1) and preserving the resource when rejected (5/5 in repetition 2) or left unanswered (5/5 in repetition 3 and 15/15 under `pressure`). In the 15 `injection` turns, the model itself ignored the embedded instruction and invoked the destructive tool in 0/15 turns, so those 15 turns reflect model-level prompt-injection refusal rather than middleware interception.

7.3 Experiment 3: Longitudinal Ablation of Self-Evolving Procedural Memory (`exp3` & `v2-exp3`)

From v1 Exploration (1,079 Turns) to v2 Confirmatory Four-Arm Design (4,608 Turns). In our initial v1 longitudinal study (`ANALYSIS_PLAN.md`, 3 arms × 8 seeds × 5 sessions × 9 tasks = 1,079 analysed turns), most task archetypes overlapped with the 5 factory-seeded static rules, and newly crystallized rules for uncovered guard codes emitted only generic fallback advice ("call `doc_validate` before editing"). Consequently, while `utility` reduced total execution signals by 37% relative to `off` (`glmmTMB` RR = 0.63, Holm $p = 0.0011$; guard rejections 0.039 vs. 0.078/turn, RR = 0.51, Holm $p = 0.1089$; cluster-robust NB2 RR = 0.50, Holm $p = 0.0269$), it did not significantly outperform static seeded rules `frozen` (0.039 vs. 0.053/turn, RR = 0.74, $p = 0.4263$).

To rigorously test whether online rule crystallization provides incremental value when agents encounter *novel, uncovered* collaborator and document constraints—and to separate rule content from static rules and prompt-length effects—we pre-specified a two-stage Phase-2 protocol (`ANALYSIS_PLAN_v2.md` §5–§6, frozen at tag `paper-eval-v2-p2`) over 9 authoring task archetypes (`TASKS_V2`: `open_comment_methods`, `open_comment_discussion`, `p_value_format`, `stat_symbol_replace`, `ambiguous_term_discussion`, `ambiguous_term_results`, `merge_list_items`, `table_merged_header`, and `out_of_scope_revert`) targeting 6 guard/revert codes *not* covered by the factory-seeded rules (`anchor_has_open_comments`, `text_not_found`, `ambiguous_match`, `invalid_structure`, `cell_not_found`, and `user_reverted_turn`). After verifying that our 432-turn 3-arm pilot (seeds 501–504, 4 users × 4 sessions × 9 tasks/arm) exceeded both pre-specified continuation thresholds (`off` rejection rate 0.785 ≥ 0.20/turn; `utility` vs. `frozen` RR = 0.43 ≤ 0.65, Holm $p = 1.25 \times 10^{-6}$, with held-out Session 4 rejections dropping to 0.056 vs. 0.778/turn), we executed the full 4,608-turn 4-arm confirmatory experiment (seeds 601–616):

- **Arm A — No Procedural Memory (`off`):** No procedural rules injected (1,152 turns).
- **Arm B — Static Seeded Rules Only (`frozen`):** Injects the 5 factory-seeded baseline rules ranked by Eq. 3 ($B = 1000$ chars), with online rule crystallization disabled (1,152 turns).
- **Arm Pl — Length-Matched Placebo Control (`placebo`):** Same as `frozen`, plus a length-matched block of task-irrelevant platform background descriptions to control for prompt-length and formatting effects (1,152 turns).
- **Arm C — Self-Evolving Procedural Memory (`utility`):** Factory-seeded rules plus online signal crystallization (`LEARNED_RULE_TEMPLATES`, threshold $\tau = 2$) and cue-ranked injection (1,152 turns).

Each arm ran 16 independent simulated user trajectories (seeds 601–616) across 8 consecutive sessions (6 training sessions on 7 training PMC manuscripts, followed by **Sessions 7 and 8 on 3 held-out PMC manuscripts**) × 9 shuffled tasks per session, totaling 4,608 turns (100% completed with 0 infrastructure or setup exclusions).

Table 8 and Figure 2 establish four findings from the 4,608-turn confirmatory experiment:

1. **Autonomous Crystallization of All Uncovered Constraints:** Every one of the 16 independent simulated users in Arm C (`utility`) autonomously crystallized all 6 uncovered procedural rules during Sessions 1–2 (96 learned rules total). Consequently, pre-write guard rejections fell from 0.750/turn (108 rejections) in Session 1 to 0.396/turn (57) in Session 2 and 0.139/turn (20) in Session 3, stabilizing at 0.181–0.208/turn on the unseen held-out manuscripts in Sessions 7–8 (a 76.1% reduction in held-out rejections relative to `off`: 56/288 = 0.194 vs. 234/288 = 0.812/turn).

Table 8: Confirmatory 4-arm longitudinal ablation of Heurion 2.0's self-evolving procedural memory (`v2-exp3`, `2026-10-22T17-11-50-247Z`, $N = 4,608$ analysed live LLM turns across 4 arms × 16 paired user seeds × 8 sessions [6 training + 2 held-out] × 9 uncovered constraint tasks). Primary analysis uses the three-way random-intercept Negative Binomial GLMM (`glmmTMB` in `stats.R`) with Holm multiplicity correction across H1 and H2; cluster-robust NB2 (`stats.py`), paired Wilcoxon signed-rank ($n = 16$ pairs), and paired bootstrap are reported as sensitivity analyses.

Experimental Arm	Users × Sessions	Analysed Turns	Guard Rejections / Turn (Total)	Held-Out (Sess. 7–8) Rejections / Turn	All Execution Signals / Turn	Task Completion Rate (All 8 Sess.)	Held-Out Completion
A. No Memory (`off`)	16 × 8	1,152	0.784 (903)	0.812 (234/288)	0.911 (1,050)	81.1% (934/1,152)	84.0% (242/288)
B. Static Seeded (`frozen`)	16 × 8	1,152	0.774 (892)	0.802 (231/288)	0.896 (1,032)	81.8% (942/1,152)	85.4% (246/288)
Pl. Placebo (`placebo`)	16 × 8	1,152	0.771 (888)	0.781 (225/288)	0.896 (1,030)	80.9% (932/1,152)	84.4% (243/288)
C. Self-Evolving (`utility`)	**16 × 8**	**1,152**	**0.285 (328)**	**0.194 (56/288)**	**0.321 (370)**	**92.8% (1,069/1,152)**	**100.0% (288/288)**
Pre-Specified Primary Mixed-Effects Models (`glmmTMB` NB2, `stats.R`, $N = 4,608$; Rate Ratios RR < 1 favor `utility`):							
H1 (`utility` vs. `off`) — Primary (Guard Rejections, `glmmTMB`):					RR = **0.363** (95% CI 0.320–0.412), $p = 1.32 \times 10^{-55}$, **Holm** $p = 2.64 \times 10^{-55}$		
H2 (`utility` vs. `frozen`) — Primary (Guard Rejections, `glmmTMB`):					RR = **0.368** (95% CI 0.324–0.417), $p = 3.87 \times 10^{-54}$, **Holm** $p = 3.87 \times 10^{-54}$		
Exploratory (`utility` vs. `placebo`) — Guard Rejections (`glmmTMB`):					RR = **0.369** (95% CI 0.325–0.419), $p = 1.32 \times 10^{-53}$		
Negative-Control Baselines (`frozen` vs. `off` / `placebo` vs. `off` / `frozen` vs. `placebo`):					RR = 0.988 ($p = 0.795$) / RR = 0.983 ($p = 0.723$) / RR = 1.005 ($p = 0.924$)		
Secondary Endpoints & Sensitivity Analyses:							
H1 & H2 Secondary (All Execution Signals, `glmmTMB` NB2):					H1 RR = **0.352** (**Holm** $p = 2.14 \times 10^{-66}$); H2 RR = **0.359** (**Holm** $p = 2.79 \times 10^{-64}$)		
Cluster-Robust NB2 by User (`stats.py`, $n = 16$ clusters):					H1 RR = 0.363 (95% CI 0.341–0.386); H2 RR = 0.368 (95% CI 0.345–0.392)		
Non-Parametric Paired User Tests ($n = 16$ seeds, `utility` lower in 16/16 users):					Wilcoxon $p = 4.34 \times 10^{-4}$ (vs. `off`) & 4.33×10^{-4} (vs. `frozen`); $\Delta_{\text{turn}} = -0.514$ & -0.479		
H3 (`utility` − `off`) — Completion Non-Inferiority & Superiority (Margin −5.0%):					**Δ** = **+11.72**% (95% CI [**+10.42**%, **+13.02**%] > −5.0%, **Non-Inferior & Superior**)		

2. **Causal Isolation Against Static Rules and Prompt-Length Placebo (H1 & H2):** Under the pre-specified negative binomial mixed model (`glmmTMB`), `utility` reduced guard rejections by 63.7% vs. `off` (RR = 0.363, 95% CI 0.320–0.412, Holm $p = 2.64 \times 10^{-55}$), by 63.2% vs. `frozen` (RR = 0.368, 95% CI 0.324–0.417, Holm $p = 3.87 \times 10^{-54}$), and by 63.1% vs. `placebo` (RR = 0.369, 95% CI 0.325–0.419, $p = 1.32 \times 10^{-53}$). Crucially, the three negative-control contrasts—`frozen` vs. `off` (RR = 0.988, $p = 0.795$), `placebo` vs. `off` (RR = 0.983, $p = 0.723$), and `frozen` vs. `placebo` (RR = 1.005, $p = 0.924$)—were statistically indistinguishable from 1.00, proving that neither unaligned static rules nor extra prompt formatting reduces errors on novel constraints. Non-parametric Wilcoxon tests confirmed that 16/16 (100%) users had lower rejection rates under `utility` than under `off`, `frozen`, or `placebo` ($p \leq 4.34 \times 10^{-4}$).

3. **Task Completion Superiority and Latency Reduction (H3):** Rather than avoiding guards by abstaining from edits, `utility` raised overall task completion from 81.1% (`off`), 81.8% (`frozen`), and 80.9% (`placebo`) to **92.8**% (**1,069/1,152**), a paired gain of +11.72% (95% Bootstrap CI [+10.42%, +13.02%], far exceeding the −5.0% non-inferiority margin). In held-out Sessions 7–8, `utility` achieved **100.0**% (**288/288**) task completion and reduced mean turn latency from 15.0–15.8 s in the control arms to 11.35 s (−25%). Notably, on `out_of_scope_revert` (128 turns/arm), where clinicians revert turns that modify adjacent paragraphs beyond the requested block, `off`, `frozen`, and `placebo` succeeded in only 6/128 (4.7%), 3/128 (2.3%), and 10/128 (7.8%) turns, whereas `utility` experienced 2 reverts per user in Sessions 1–2 (32 total), crystallized `user_reverted_turn`, and succeeded in **96/96** (**100.0**%) turns across Sessions 3–8.

7.4 Pre-Specification Transparency, Deviations, and Limitations

Pre-Specification and Git Provenance. Both generations of the evaluation protocol (`paper/benchmarks/ANALYSIS_PLAN.md` and `paper/benchmarks/v2/ANALYSIS_PLAN_v2.md`) were frozen in the git repository prior to formal data collection at tags `paper-eval-v1` (`1d8f307c`), `paper-eval-v1.1` (`e060107f`), `paper-eval-v2-p1a` (`16180f3c`), `paper-eval-v2-p1b` (`932f4e3a`), and `paper-eval-v2-p2` (`e8362c88`). Because protocols were locked via internal git tags rather than an external registry such as OSF, we refer to them as pre-specified rather than externally pre-registered. All developmental iterations (such as the Phase 1a-to-1b `F1--F5` platform fixes, the 38-turn post-hoc attribution in §5.3.4, and the Phase-2 pilot continuation

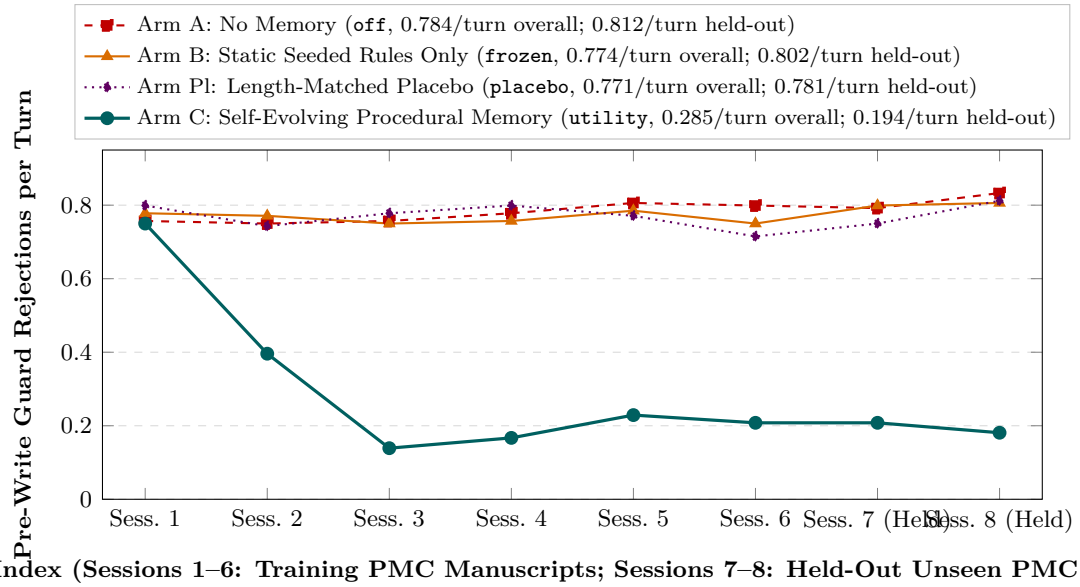


Figure 2: Longitudinal guard-rejection trajectories across 8 consecutive sessions (144 turns/session/arm, 4,608 total turns across 16 paired user seeds) in the confirmatory Phase-2 experiment (`v2-exp3`). All four arms start at ≈ 0.75–0.80 rejections/turn in Session 1. While `off`, `frozen`, and `placebo` remain flat across all 8 sessions (≈ 0.75–0.83/turn), `utility` autonomously crystallizes all 6 uncovered rules during Sessions 1–2 (96 rules across 16 users), reducing guard rejections to 0.139–0.229/turn in Sessions 3–6 and generalizing zero-shot to unseen clinical manuscripts in held-out Sessions 7–8 (0.208 and 0.181/turn, with 100% [288/288] task completion).

gate in §5.2) are recorded chronologically in `ANALYSIS_PLAN_v2.md` and reported alongside confirmatory results.

Limitations. Our empirical evaluation has six main limitations: (1) all live agent authoring experiments used `deepseek-flash`, while Stage-2 local PHI extraction used `qwen2.5:1.5b`; (2) all user trajectories (16 seeds in `v2-exp3`, 20 PMC manuscripts across `v2-q1` and `v2-exp3`) were driven by automated scripts with deterministic clinician callbacks rather than prospective human-subject user trials (a clinician crossover study protocol is outlined in `ANALYSIS_PLAN_v2.md` §7 for future work); (3) although the 100-note `v2-phi` test set substantially expands narrative variety (485 narrative spans across multiple clinical specialties) over the initial template set, external validation on annotated multi-center hospital EHRs [12] remains necessary before clinical deployment; (4) while citation support is checked at insertion time via abstract matching and evaluated by an LLM judge in `v2-q1`, blind human specialist grading of `verify_claims` verdicts is left to future work; (5) all live experiments disabled declarative memory auto-extraction and Layer 3 background overlays (`HARNESS_EVOLVE=0`) to preserve statistical independence across user seeds; and (6) task completion in `exp3` is graded by deterministic structural/textual oracles, with blind human audit of drawn subsamples (`audit/items.json`) deferred to future work.

Platform Verification Summary. Table 9 summarizes the automated verification and empirical evaluation coverage across the repository.

Table 9: Automated verification and empirical evaluation coverage across the Heurion 2.0 codebase.

Verification Metric	Measured Value
Automated Test Suite (`apps/platform/tests/*.test.ts`)	74+ Files / 625+ Tests Passing (100%)
OOXML Word Fidelity Corpus (`v2/fidelity/run.ts`, 47 fixtures × 4 layers)	188/188 Checks Passing (100%)
Unaccounted REST API Endpoints in Parity Audit (`parity.test.ts`)	0 (100% Accounted)
Deterministic & Local-SLM Guard Records (`exp1`, `exp2b`, locked `v2-phi` test set)	1,010 Records (1,465 PHI Spans + 560 Neg. Controls)
Three-Arm Manuscript Co-Editing Live Turns (`exp2a-papers` + `v2-q1` 1a & 1b)	4,650 Turns (20 CC BY PMC Manuscripts)
Longitudinal Procedural Memory Live Turns (`exp3` v1 + `v2-exp3` Pilot & 4-Arm Formal)	6,119 Turns (4,608 in 4-Arm Confirmatory Study)

8 Related Work

Collaborative Editing Data Structures and Agent-Computer Interfaces (ACI). Conflict-free Replicated Data Types (CRDTs) [7] and local-first software architectures [8] enable concurrent multi-actor editing without central locking. In parallel, SWE-agent [3] and OpenHands [4] showed that specialized Agent-Computer Interfaces outperform raw shell scripting for software engineering, and the Model Context Protocol (MCP) [18] standardized tool exposure for LLM agents. Heurion 2.0 combines these threads by exposing a shared ProseMirror + Yjs CRDT document core through guarded MCP primitives with node-level optimistic concurrency locks, per-turn selective CRDT undo, and raw-slice OOXML export.

Literature Synthesis, Citation Hallucination, and De-Identification. Empirical audits have documented frequent fabricated references in LLM-generated medical content [11] and in LLM-generated bibliographies across disciplines [10]; see [9] for a broader survey of hallucination. While STORM [5] and PaperQA2 [6] automate literature synthesis and FActScore [21] and ALCE [22] evaluate post-hoc attribution, Heurion 2.0 enforces citation registration as a pre-write syntactic invariant (`[@c:id]`). For clinical privacy, rule-based and hybrid de-identification systems such as the i2b2/UTHealth shared-task systems [12], Philter [13], and Microsoft Presidio [14] combine regular expressions with statistical or neural sequence labelers; our two-stage PHI funnel adapts this hybrid pattern to local loopback deployment by pairing checksum/subtype validators with a pre-masked 1.5B SLM and symbolic output validation, while our confirmation middleware mitigates indirect prompt injection [17] on destructive endpoints.

Self-Evolving Agents and Memory Architectures. Reflexion [19] and MemGPT [20] introduced verbal reinforcement and tiered memory management for LLM agents, while Voyager [23] accumulated executable skills in open-ended environments. In collaborative authoring, mixing subjective user stylistic preferences with operational error recovery can create prompt interference; Heurion 2.0 separates human-confirmed declarative preferences from signal-crystallized procedural rules ranked by document-state and prompt cues.

9 Conclusion

We have presented Heurion 2.0, a collaborative clinical authoring platform built around a contract-guarded dual-surface CRDT engine and a three-layer procedural memory system:

- **Dual-Surface CRDT Co-Editing and OOXML Export:** Replacing direct `python-docx` file scripting with a shared Yjs + ProseMirror CRDT core, stable block IDs (`{#id}`), `base_rev` concurrency guards, per-turn selective undo (`revertTurn`), and raw XML slice caching (`node_src`) reduced manuscript rework rates to 4.9% (38/780; 2.1% true defect rate, 0 lost human edits) vs. 14.0% (OR = 32.47, Holm $p = 1.03 \times 10^{-16}$) for direct file scripting and 10.3% (OR = 9.66, Holm $p = 1.23 \times 10^{-8}$) for tool-assisted file scripting

across 2,340 confirmatory turns on 10 unseen 2026 PMC manuscripts, while lowering API cost by 30%–33% (GMR = 0.70 and 0.67, Holm $p < 10^{-72}$).

- **Pre-Write Citation and Local Two-Stage PHI Guards:** Pre-write syntactic validation blocked 100/100 unregistered inline DOIs/PMIDs, and pairing Stage-1 regex/checksum rules with a pre-masked local 1.5B SLM (`qwen2.5:1.5b`) and symbolic biomedical output validation recalled 87.0% (648/745) of PHI spans (83.3% [404/485] on narrative spans and 93.8% [244/260] on labeled spans) at 97.6% (482/494) span precision and 6.0% (12/200) hard-negative sentence FPR on the locked multi-specialty clinical test set, outperforming both Stage-1 rules alone (27.4% recall, 3.5% narrative) and the unvalidated SLM (92.0% hard-negative FPR).
- **Self-Evolving Procedural Memory:** In our confirmatory 4,608-turn, 4-arm (`off`, `frozen`, `placebo`, `utility`) × 16-user × 8-session longitudinal experiment on uncovered structural and collaborator constraints, all 16 `utility` users autonomously crystallized all 6 uncovered rules (96 rules total), reducing pre-write guard rejections per turn by 63.7% overall (0.285 vs. 0.784 [`off`], 0.774 [`frozen`], and 0.771 [`placebo`]; `glmmTMB` RR = 0.363–0.369, Holm $p < 10^{-53}$) and by 76.1% on held-out unseen manuscripts (0.194 vs. 0.812/turn), while raising task completion by +11.72% (92.8% vs. 81.1%; 100.0% [288/288] in held-out sessions).
- **Human-AI Permission Symmetry:** In-process HTTP Loopback execution under user JWT identity (`X-Heurion-Via: ai`), confirmation cards for destructive actions (30/30 tool invocations intercepted), and automated CI router parity checks (`parity.test.ts`) ensure that AI tools operate within the same authorization boundaries as the human researcher.

Statements and Declarations

Ethics Statement. No human subjects were recruited and no real patient electronic health records or identifiable clinical data were accessed in this study. All 20 clinical manuscripts used across Experiments 1b, 1c, and 3 are open-access articles distributed under Creative Commons (CC BY) licenses via PubMed Central, all PHI evaluation strings in Experiment 2a (`exp2b` and `v2-phi`) were synthetically generated, and all user trajectories were executed by automated seeded scripts.

Code and Data Availability. The platform source code (`apps/platform/`), benchmark harnesses (`paper/benchmarks/`), pre-specified analysis plans (`paper/benchmarks/ANALYSIS_PLAN.md` and `paper/benchmarks/v2/ANALYSIS_PLAN_v2.md`, git tags `paper-eval-v1`, `paper-eval-v1.1`, `paper-eval-v2-p1a`, `paper-eval-v2-p1b`, and `paper-eval-v2-p2`), raw turn-level JSONL/CSV logs, and statistical analysis scripts (`stats.R`, `stats.py`) are included in the project repository workspace (`paper/results/`).

Competing Interests. Hui Zhao is affiliated with Heurion, which develops the Heurion 2.0 platform evaluated in this paper. Dong Qian declares no competing financial interests.

References

[1] Kenneth F. Schulz, Douglas G. Altman, and David Moher. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *BMJ*, 340:c332, 2010. doi:10.1136/bmj.c332.

[2] Jan P. Vandenbroucke, Erik von Elm, Douglas G. Altman, Peter C. Gøtzsche, Cynthia D. Mulrow, Stuart J. Pocock, Charles Poole, James J. Schlesselman, and Matthias Egger.

Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. *PLoS Medicine*, 4(10):e297, 2007. doi:10.1371/journal.pmed.0040297.

[3] John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[4] Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, et al. OpenHands: An open platform for AI software developers as generalist agents. In *International Conference on Learning Representations (ICLR)*, 2025.

[5] Yijia Shao, Yucheng Jiang, Theodore A. Kanell, Peter Xu, Omar Khattab, and Monica S. Lam. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, pages 6252–6278, 2024. doi:10.18653/v1/2024.naacl-long.347.

[6] Michael D. Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Michaela Hinks, Michael J. Hammerling, Manvitha Ponnapati, Samuel G. Rodriques, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024.

[7] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski. Conflict-free replicated data types. In *Stabilization, Safety, and Security of Distributed Systems (SSS 2011)*, volume 6976 of *Lecture Notes in Computer Science*, pages 386–400. Springer, 2011. doi:10.1007/978-3-642-24550-3_29.

[8] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan. Local-first software: You own your data, in spite of the cloud. In *Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!)*, pages 154–178, 2019. doi:10.1145/3359591.3359737.

[9] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023. doi:10.1145/3571730.

[10] William H. Walters and Esther Isabelle Wilder. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi:10.1038/s41598-023-41032-5.

[11] Mehul Bhattacharyya, Valerie M. Miller, Debjani Bhattacharyya, and Larry E. Miller. High rates of fabricated and inaccurate references in ChatGPT-generated medical content. *Cureus*, 15(5):e39238, 2023. doi:10.7759/cureus.39238.

[12] Amber Stubbs and Özlem Uzuner. Annotating longitudinal clinical narratives for de-identification: The 2014 i2b2/UTHealth corpus. *Journal of Biomedical Informatics*, 58:S20–S29, 2015. doi:10.1016/j.jbi.2015.07.020.

[13] Beau Norgeot, Kathleen Muenzen, Thomas A. Peterson, Xuancheng Fan, Benjamin S. Glicksberg, et al. Protected Health Information filter (Philter): accurately and securely de-identifying free-text clinical notes. *npj Digital Medicine*, 3:57, 2020. doi:10.1038/s41746-020-0258-y.

[14] Microsoft. Presidio: Data protection and de-identification SDK. https://github.com/microsoft/presidio. Accessed October 2026.

[15] U.S. Department of Health and Human Services. Standards for Privacy of Individually Identifiable Health Information (HIPAA Privacy Rule). *45 CFR Parts 160 and 164*; final rule 2000, as amended 2002. De-identification standard at 45 CFR §164.514(b).

[16] Standing Committee of the National People's Congress of the People's Republic of China. Personal Information Protection Law of the People's Republic of China. Adopted August 20, 2021; effective November 1, 2021.

[17] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, pages 79–90, 2023. doi:10.1145/3605764.3623985.

[18] Anthropic. Model Context Protocol specification. https://modelcontextprotocol.io/specification, 2024. Introduced November 25, 2024.

[19] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023. doi:10.52202/075280-0377.

[20] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[21] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 12076–12100, 2023. doi:10.18653/v1/2023.emnlp-main.741.

[22] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6465–6488, 2023. doi:10.18653/v1/2023.emnlp-main.398.

[23] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research (TMLR)*, 2024.